# The dog game: policy-gradient and Nash-Q results

Convergence to the analytical equilibrium, and the angle-radius (walking) version

## Summary


- Policy-gradient self-play reached exploitability below 0.004 in 74 of 75 runs (5 house setups x 3 network architectures x 5 seeds, 1500 episodes each).
- The miss is: separate architecture, house setup 2, seed 0 (0.18). I have not tested why it is slower or whether more training would fix it.
- The three architectures do not separate: separate 24 of 25, shared 25 of 25, partial 25 of 25.
- In the angle-radius game, both REINFORCE and the 10-direction Nash-Q network move the players toward the one-shot Nash corners (mean distance 1.49 to 0.04 and 1.95 to 0.01).
- With A2C and PPO added to the walking game, A2C is the most reliable (5 of 5 seeds at the corners), and PPO is the fastest to get there but the least stable (2 of 5 seeds at the end).


## The game

Two houses sit at fixed points in the unit square. Each round, both players pick a point and the dog moves to the weighted average w * red + (1 - w) * blue. Each player is rewarded by the negative squared distance from the dog to their own house, with discount 0.9. Because the domain is bounded, players overshoot toward the edge to counteract each other. The one-shot equilibrium is found by iterated best response (doggame/nash.py), and a learned policy is scored by its exploitability: how much a player could still gain by deviating, searched on a 101 x 101 grid.

## Does policy gradient converge?

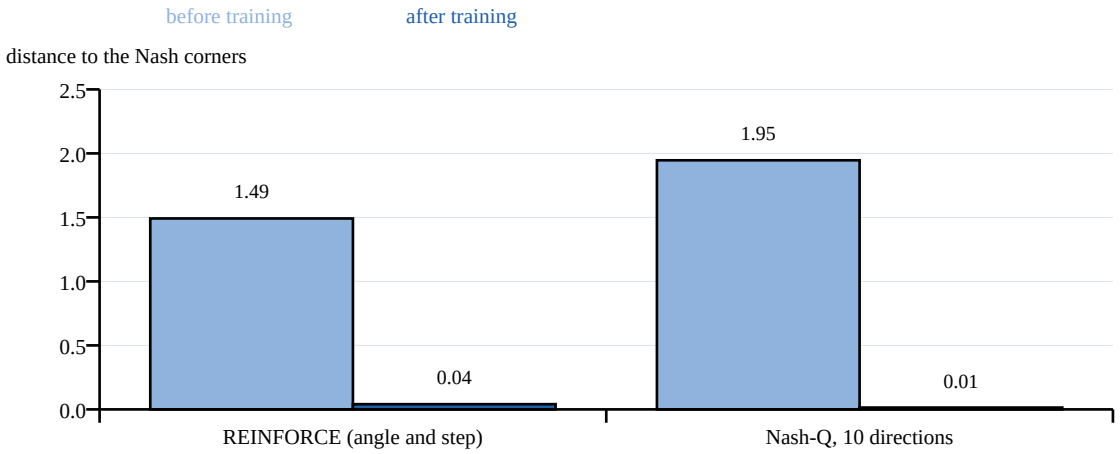
REINFORCE self-play, 5 seeds (0-4) for each setup and architecture. Each cell is the number of runs under 0.004, with the median exploitability.

| House setup | separate | shared | partial |
|---|---|---|---|
| red (0.9, 0.2), blue (0.1, 0.8), w = 0.5 | 5 of 5 (median 0.0002) | 5 of 5 (median 0.0002) | 5 of 5 (median 0.0004) |
| red (0.8, 0.8), blue (0.2, 0.3), w = 0.3 | 4 of 5 (median 0.0007) | 5 of 5 (median 0.0006) | 5 of 5 (median 0.0008) |
| red (0.7, 0.1), blue (0.6, 0.9), w = 0.7 | 5 of 5 (median 0.0003) | 5 of 5 (median 0.0003) | 5 of 5 (median 0.0008) |
| red (0.95, 0.5), blue (0.05, 0.5), w = 0.5 | 5 of 5 (median 0.0001) | 5 of 5 (median 0.0001) | 5 of 5 (median 0.0002) |
| red (0.6, 0.6), blue (0.4, 0.4), w = 0.5 | 5 of 5 (median 0.0001) | 5 of 5 (median 0.0001) | 5 of 5 (median 0.0002) |

- The 0.004 threshold is my choice. The closest passing run is 0.0039 (shared architecture, house setup 2, seed 0), so the count of passing runs depends on where that line sits.
- House setup 4 has both houses at y = 0.5, so the y-axis of the stage game is degenerate and many y-picks are equally good. All 15 runs there have exploitability of at most 0.0002, so whichever equilibrium they found is not exploitable.

## Angle-radius version: walking instead of jumping

Each player picks an angle and a step length (up to 0.2) and walks from where it stands. The score is the sum of both players' distances from the one-shot Nash corners at the end of a game, averaged over 10 sampled games, on one house setup (red (0.88, 0.12), blue (0.3, 0.65), w = 0.5). REINFORCE trains for 300 episodes and the Nash-Q network for 60, over 5 seeds.




Mean distance over the seeds, before and after training.

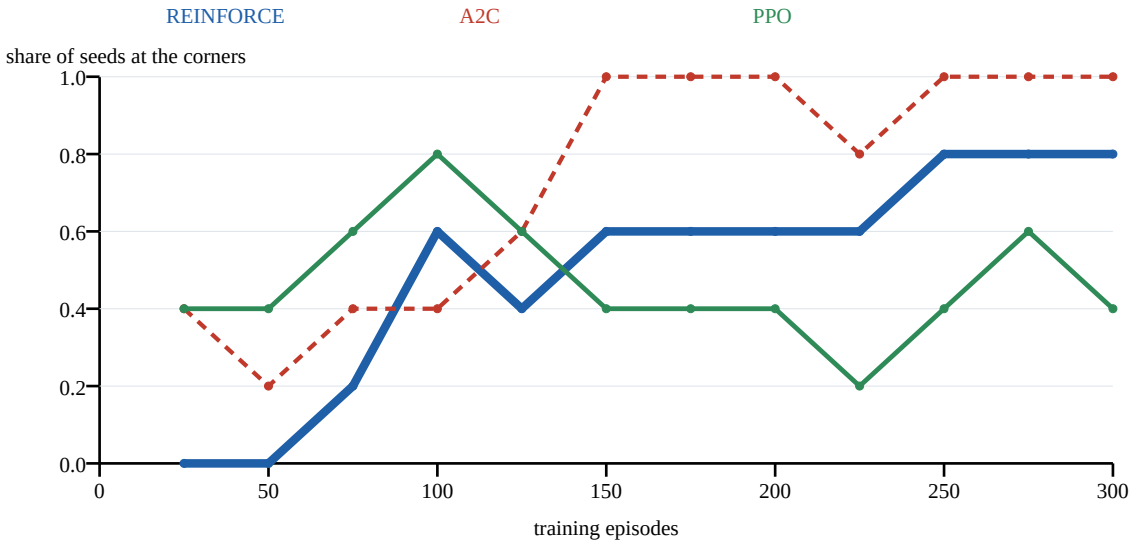
| Method | Distance before training | Distance after training | Seeds that improved |
|---|---|---|---|
| REINFORCE (angle and step) | 1.49 | 0.04 | 5 of 5 |
| Nash-Q, 10 directions | 1.95 | 0.01 | 5 of 5 |

Per seed:

| Seed | REINFORCE (angle and step): before / after | Nash-Q, 10 directions: before / after |
|---|---|---|
| 0 | 1.72 / 0.05 | 2.18 / 0.07 |
| 1 | 2.09 / 0.00 | 1.67 / 0.00 |
| 2 | 1.87 / 0.05 | 1.41 / 0.00 |
| 3 | 1.66 / 0.07 | 2.05 / 0.00 |
| 4 | 0.12 / 0.03 | 2.41 / 0.00 |

## REINFORCE, A2C and PPO on the walking game

The same three algorithms on the walking game, five seeds each, 300 episodes, scored every 25 episodes. "At N" is the mean distance after N episodes. A seed is at the corners when the summed distance is under 0.1 (my threshold). The chart shows the share of seeds at the corners.




| Algorithm | Before training | At 50 | At 100 | At 200 | At 300 | Late checkpoints at the corners (from 150) | Seeds at the corners at the end |
|---|---|---|---|---|---|---|---|
| REINFORCE | 1.49 | 0.696 | 0.341 | 0.130 | 0.055 | 24 of 35 | 4 of 5 |
| A2C | 1.49 | 0.731 | 0.199 | 0.032 | 0.011 | 34 of 35 | 5 of 5 |
| PPO | 1.49 | 0.263 | 0.127 | 0.500 | 0.371 | 14 of 35 | 2 of 5 |

- A2C is the most reliable: 5 of 5 seeds at the corners at the end, and 34 of 35 later checkpoints.
- REINFORCE is slower and more variable (4 of 5 seeds, 24 of 35 later checkpoints).
- PPO is the fastest to get there but does not stay: 2 of 5 seeds at the end and 14 of 35 later checkpoints. This is the opposite of the main game, where PPO was the fastest and most reliable. Why is not tested.

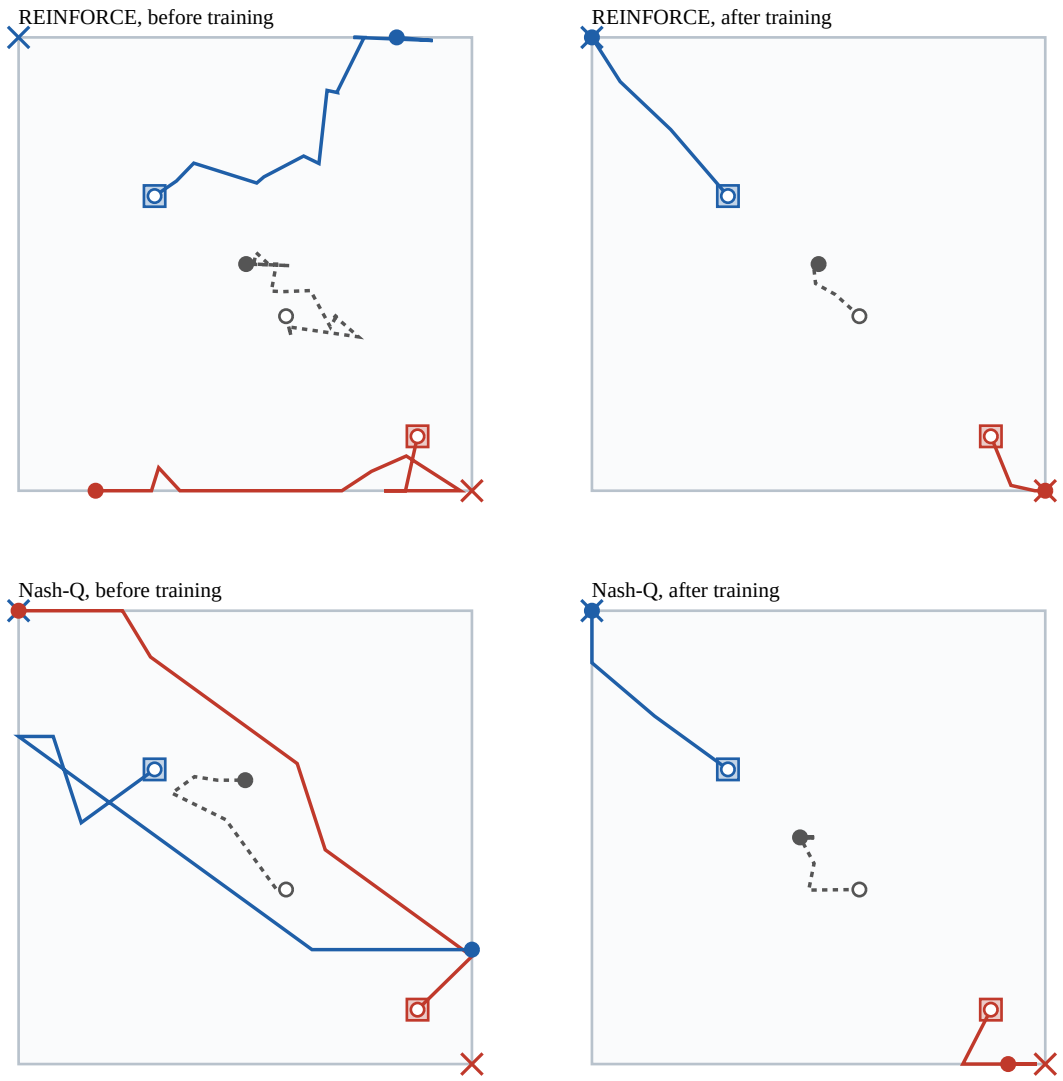
Can PPO be made reliable on the walking game? PPO with one epoch and/or a smaller learning rate, five seeds each, scored the same way:

| Setting | Before training | At 50 | At 100 | At 200 | At 300 | Late checkpoints at the corners (from 150) | Seeds at the corners at the end |
|---|---|---|---|---|---|---|---|
| A2C (reference) | 1.49 | 0.731 | 0.199 | 0.032 | 0.011 | 34 of 35 | 5 of 5 |
| PPO, lr 0.01, 4 epochs (default) | 1.49 | 0.263 | 0.127 | 0.500 | 0.371 | 14 of 35 | 2 of 5 |
| PPO, lr 0.01, 1 epoch | 1.49 | 0.595 | 0.479 | 0.249 | 0.123 | 21 of 35 | 4 of 5 |
| PPO, lr 0.003, 4 epochs | 1.49 | 0.477 | 0.361 | 0.448 | 0.362 | 15 of 35 | 1 of 5 |
| PPO, lr 0.003, 1 epoch | 1.49 | 0.836 | 0.641 | 0.470 | 0.322 | 8 of 35 | 2 of 5 |

- Fewer epochs helps partly (4 of 5 seeds at the corners, against 2 of 5 for the default), a smaller learning rate does not, and nothing matches A2C (5 of 5). Why is not tested.

## Trajectories

One sampled 15-step game for seed 0, before and after training, on the same house setup. Red and blue walk from their own houses; the dog moves to the average of their positions.




Paths of red, blue and the dog. Before training the walks wander; after training both players end at or near their Nash corners and the dog settles near the middle.

## Limits

- The corners are the equilibrium of the one-shot game. The walking game itself has not been solved, so this shows the players heading to the one-shot corners, not that they play an equilibrium of the walking game.
- Walking was checked on one house setup only. The Nash-Q network sees the exact stage-game solution at every target, so it is not independent of the solver.
- The untrained REINFORCE (angle and step) policy of seed 4 already started close to the corners (0.12), so that run shows little.

- 5 seeds, fixed training lengths and learning rates (0.01 for REINFORCE, 0.001 for Nash-Q), none tuned for these runs.
- The reading of the angle-radius action as a move from the player's own position, and of the dog weights, are open questions (see ASSUMPTIONS.md).

Raw numbers are in results/dog_game_runs.jsonl. The runs are doggame/dog_game_study.py.